

Callable Bermudan Cross-Currency Swap - Quantitative Model Summary

Model: two Hull-White rate factors + lognormal FX + Bermudan LSM
Numerator: domestic money-market account; values in domestic currency
Perspective: positive cash flows are receipts by the exercise holder

1. Product and notation

Consider a two-currency swap with domestic and foreign cash-flow legs, optional initial/final notional exchanges, and exercise dates $t_1 < \dots < t_M$. Define

- S_t as FX in units of DOM per one unit of FOR;
- $P_d(t, T)$ as the domestic collateral discount bond; and
- $P_f^*(t, T)$ as the foreign effective discount bond consistent with domestic collateral and cross-currency basis.

A foreign cash flow $CF_f(T)$ has pathwise domestic value $S_T CF_f(T)$. All cash flows are discounted by the domestic money-market numeraire.

At decision date t_i , partition the value into common cash flows A_i , an exercise-only settlement J_i , and the cancellable continuation tail Y_i . With continuation estimate

$$C_i = \mathbb{E}^{\mathbb{Q}^d}[Y_i | \mathcal{F}_{t_i}], \quad \tau^* = \inf\{t_i : J_i > C_i + \varepsilon_{\text{ex}}\},$$

the decision-date value is $A_i + \max(J_i, C_i)$. For zero-fee cancellation, $J_i = 0$, so the holder cancels when continuation is negative. For an entry option, J_i is the state-dependent value of entering the swap plus settlement.

2. Joint stochastic model

Work under the domestic risk-neutral measure $\mathbb{Q}^d$. With $S = \text{DOM}/\text{FOR}$, use Hull-White one-factor rates and lognormal FX:

$$dr_d(t) = [\theta_d(t) - a_d r_d(t)] dt + \sigma_d(t) dW_d(t), \quad (1)$$

$$dr_f(t) = [\theta_f(t) - a_f r_f(t) - \rho_{fS} \sigma_f(t) \sigma_S(t)] dt + \sigma_f(t) dW_f(t), \quad (2)$$

$$\frac{dS_t}{S_t} = [r_d(t) - r_f(t)] dt + \sigma_S(t) dW_S(t). \quad (3)$$

The negative foreign-rate quanto term is the measure-change adjustment for the stated FX quotation. Calibrate the foreign curve fit under its foreign/effective numeraire measure and then transform to $\mathbb{Q}^d$. Equivalently, verify

$$S_0 P_f^*(0, T) = \mathbb{E}^{\mathbb{Q}^d}[D_d(0, T) S_T].$$

Reversing the FX quotation or changing numeraire changes the adjustment and requires a fresh derivation.

The instantaneous Brownian correlation matrix is

$$R = \begin{pmatrix} 1 & \rho_{df} & \rho_{dS} \\ \rho_{df} & 1 & \rho_{fS} \\ \rho_{dS} & \rho_{fS} & 1 \end{pmatrix}, \quad R = R^T, \quad R \succeq 0.$$

Constant correlation is the base model. FX is Black-Scholes only in diffusion form; stochastic rates make the full hybrid model non-analytic.

Multi-curve assumption

Use one stochastic rate factor per currency and deterministic forecast/discount basis. For currency c , freeze

$$B_c(0, T) = \frac{P_c^F(0, T)}{P_c^D(0, T)}, \quad P_c^F(t, T) = P_c^D(t, T) \frac{B_c(0, T)}{B_c(0, t)}.$$

Forward coupons follow from $P_c^F(t, T)$. Stochastic tenor or cross-currency basis requires another factor and is outside the base model.

3. Calibration

- **Curves:** bootstrap domestic collateral and foreign effective curves from liquid OIS/basis instruments. Reprice the calibration set and enforce FX-forward parity:

$$F_{FX}(0, T) = S_0 \frac{P_f^*(0, T)}{P_d(0, T)}.$$

- **Domestic HW:** calibrate to domestic normal-vol swaptions on the exercise diagonal. Fix or tightly constrain a_d ; regularize/bound piecewise $\sigma_d(t)$.
- **Foreign HW:** use the analogous foreign swaption fit and identification constraints for a_f and $\sigma_f(t)$.
- **FX diffusion:** fit piecewise $\sigma_S(t)$ to selected ATM/benchmark-tenor FX vanillas in the full hybrid. For multiple strikes, minimize weighted price errors and report smile residuals; exact smile fitting needs local/stochastic volatility.
- **Correlation:** use governed historical/implied estimates, validate positive semidefiniteness, and stress ρ_{df} , ρ_{dS} , and ρ_{fS} independently.

Retain fitted parameters, helper market/model prices, relative errors, optimizer status, and curve/vol/correlation snapshot IDs. Test parameter stability and out-of-sample repricing.

4. Monte Carlo discretization

The event grid contains every fixing, accrual, payment, notional-exchange, decision, effective, and settlement date, with intermediate steps capped at a chosen maximum interval. Generate correlated normals from a Cholesky/eigen factorization of R .

- Use exact conditional Gaussian Hull-White transitions when available.
- Integrate domestic short rates consistently for pathwise discount factors.
- Evolve log FX using

$$d \log S_t = [r_d(t) - r_f(t) - \frac{1}{2} \sigma_S^2(t)] dt + \sigma_S(t) dW_S(t).$$

- Value conditional zero-coupon bonds analytically from Hull-White states where possible and apply the deterministic basis mapping to forecast curves.
- Start with antithetics; validate Sobol/Brownian-bridge variants independently.

An independently calculated non-callable XCCY cash-flow PV is the primary control variate and deterministic benchmark.

5. Bermudan exercise by least-squares Monte Carlo

Use a two-pass Longstaff-Schwartz algorithm.

Training pass. Work backward from t_M . At t_i , regress the value discounted to t_i of cancellable cash flows in $(t_i, t_{i+1}]$ plus the optimized value at t_{i+1} . Exclude common A_i flows from both alternatives and add them outside the exercise comparison. A standardized basis may include

$$1, r_d, r_f, \log S, PV_d, SPV_f, r_d^2, r_f^2, (\log S)^2, \\ r_d r_f, r_d \log S, r_f \log S.$$

Use all alive paths for cancellation unless a candidate filter is validated. Record rank, condition number, coefficients, path count, and residual diagnostics; then freeze preprocessing and coefficients.

Pricing pass. Apply the frozen policy forward on independent paths. The first exercise stops only the cancellable tail; retain paid and delayed/common cash flows and add settlement. Discount each realized cash flow to time zero and average. The result is an LSM lower bound. Re-train across seeds and evaluate policies on one common pricing sample to separate policy dispersion from sampling noise.

6. Measures, risks, and validation

Report callable and non-callable NPV, embedded option value, Monte Carlo error/confidence interval, exercise/no-exercise probabilities, expected exercise time, per-leg PV, and calibration/regression diagnostics. From the holder perspective,

$$V_{\text{option}} = V_{\text{callable}} - V_{\text{noncallable}} \geq 0$$

up to the covariance-aware Monte Carlo error.

Risk uses bump-and-revalue with common random numbers: domestic/foreign curve DV01, forecast- and cross-currency-basis risk, rate vegas, FX delta/gamma/vega, three correlation sensitivities, and Hull-White parameter stresses. State whether each bump recalibrates the model and retraining the stopping policy.

Minimum validation:

1. no exercise dates reproduce the non-callable XCCY PV;
2. zero-vol and identical-currency limits reproduce deterministic/single-currency values;
3. one exercise date matches an independent European or nested-MC benchmark;
4. domestic-numeraire-discounted domestic and FX-converted foreign assets are martingales under $\mathbb{Q}^d$;
5. prices converge across paths, grids, bases, and training seeds;
6. representative multi-exercise cases compare the LSM lower bound with a dual upper bound or trusted nested-MC benchmark; and
7. differences use paired pathwise or covariance-aware errors under common random numbers.

References. F. A. Longstaff and E. S. Schwartz (2001), "Valuing American Options by Simulation: A Simple Least-Squares Approach." QuantLib documentation: <https://www.quantlib.org/docs.shtml>.